Jakub Jan **Duchniewicz**

SENIOR EMBEDDED AND SOFTWARE ENGINEER · TECHNICAL LEAD

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Summary

Embedded systems engineer with experience across low-level software, production systems, and hardware / software integration. My work spans Linux and operating systems, telecommunications software, FPGA- and ASIC-adjacent development, performance optimization, bring-up, and cross-layer debugging. Over the past five years, I have focused extensively on 5G and LTE systems in firmware, protocol-layer software, and real-time environments.

I am strongest where specifications are incomplete and systems are unstable: diagnosing root causes across hardware/software boundaries, making sound technical trade-offs, and turning difficult systems into shippable solutions. I have led engineers, supported production releases, worked directly with customers, and owned critical components in both commercial and open-source codebases.

Experience

Trustworthy Systems, UNSW
Sydney, Australia

OPERATING SYSTEM ENGINEER — *Oct. 2025 – Present*

- Developed drivers and board support for the Radxa Rock 3B in seL4 kernel and Microkit-based systems.
- Extended CI pipeline and regression testing for the sDDF project, improving confidence in platform support and system changes.
- Diagnosed and fixed a cross-platform cycle regression caused by incorrect FPU configuration, restoring expected performance characteristics.
- Designed, implemented, integrated, and tested a virtual switch sDDF component, allowing for a more complete system stack where virtio-based VMs can talk to each other and external world.

Tietoevry — client engagements: Intel, Sequans, United Micro (later Bridgecom)
Remote (Wrocław, Poland)

SENIOR EMBEDDED AND SOFTWARE ENGINEER — *Dec. 2021 – Oct. 2025*

- Led, supported, and mentored engineers across embedded and telecommunications projects, combining hands-on development with technical coordination and customer-facing work.
- Worked across 5G modem, O-RAN, and low-level systems software, contributing in areas including performance optimization, security hardening, fuzzing, protocol-layer development, and hardware/software debugging.
- Participated in client-facing technical discussions and bids, helping shape delivery approaches and technical scope for prospective engagements.
- Took ownership of difficult technical areas where specifications, software, and hardware behavior diverged, especially in performance-critical and integration-heavy environments.
- **Intel** (Dec. 2021 – Jul. 2022): worked on performance, security, and fuzzing coverage, finding and fixing multiple bugs in production-relevant code paths and improving confidence in release readiness.
- **Intel** (Aug. 2022 – Oct. 2023): initially developed new 5G features for FlexRAN (support for new bands, 5G SA improvements); later led a team of 7 engineers, designed the architecture and interfaces, chose programming languages, tools and collaborated closely with 2 additional teams for a year to deliver an O-RAN M-plane solution.
- **Sequans** (Nov. 2023 – Jun. 2024): designed and delivered a bare-metal modem driver and implemented blind decoding for the 5G PDCCH PHY channel.
- **Sequans** (Nov. 2023 – Jun. 2024): resolved issues at the hardware/software boundary by diagnosing unstable behavior, identifying root causes from incomplete specifications, and implementing practical software workarounds during bring-up.
- **United Micro** (Jul. 2024 – Jul. 2025): ported firmware and low-level code to a 5G eRedCap solution, implemented RSSI scan in L1 synchronization layer, designed CSI-RS handling and made system-level design decisions during bring-up with partial specifications and evolving hardware behavior.
- **United Micro** (Jul. 2024 – Jul. 2025): brought-up RSSI scan procedure from the ground up, debugged RTL and software in tandem, resolved integration issues, and improved memory layout and runtime behavior in performance-sensitive paths.

Sticky Piston Studios
Remote

CO-FOUNDER / SOFTWARE ENGINEER — *Dec. 2020 – Present*

- Co-founded a studio building products and technical prototypes at the intersection of software, embedded systems, and interactive media.
- Designed and shipped City Event Map, later deployed in the Vilnius region, Lithuania.
- Acted as an architect and maintainer for internal product codebases, making system-level technical decisions across implementation, tooling, and deployment.
- Maintain and evolve in-house projects, including open-source work on Pill Engine.

Google Summer of Code, BeagleBoard.org
Remote

EMBEDDED SOFTWARE ENGINEER — *May 2021 – Aug. 2021*

- Built a C library enabling GPGPU-style computation on BeagleBone Black using OpenGL ES 2.0 and EGL.
- Implemented reusable computation pipelines including scalar operations, array operations, and 2D convolution.
- Benchmarked performance across varying data sizes and documented the work publicly.

Samsung Electronics
Warsaw, Poland / Suwon, South Korea

JUNIOR SOFTWARE ENGINEER — *Feb. 2018 – Mar. 2020*

- Owned a production broadcast middleware component in Tizen across 3 release cycles. Working in C++, C, and ARM assembly across development, debugging, optimization, and release support.
- Improved performance and reliability in a major embedded broadcast pipeline by 20% through targeted refactoring, thread-level optimization, and careful analysis of runtime behavior.
- Supported production releases and client-facing escalations, including overseas defect handling and on-site support during commercialization in South Korea.
- Investigated coredumps, logs, and system failures to diagnose issues quickly and provide fixes or actionable guidance to dependent teams.
- Built internal tools and automation in Python and Bash to improve debugging, defect analysis, and engineering efficiency.

Skills

Languages	C, C++, Rust, Python, Bash, ARM/x86/RISC-V assembly, Verilog, SystemVerilog, VHDL
Embedded	Device drivers, bring-up, Linux kernel, seL4, Microkit, RTOS, bootloaders, microcontrollers, SoCs, ASICs, FPGA
Telecom	5G/LTE L1/L2, O-RAN, FlexRAN, DPDK, network stack, protocol implementation, fuzzing
Specialties	Performance optimization, hardware acceleration, DSP, reverse engineering, security, audio/video, game engines
Human Languages	Polish, English, German, Spanish

Education

EIT Digital Master School / KTH Royal Institute of Technology / University of Turku
Europe

M.SC. IN EMBEDDED SYSTEMS — *Sept. 2020 – Nov. 2022*

- Recipient of the EIT Excellence Scholarship.
- Thesis: FPGA-accelerated packet capture with eBPF.

Warsaw University of Technology
Warsaw, Poland

B.SC. IN COMPUTER SCIENCE AND NETWORKS — *Oct. 2016 – Aug. 2020*

- Thesis: FPGA-based hardware accelerator for musical synthesis on Linux.
- Nominated for the IEEE Polish Diploma Contest.

Open Source & Leadership

PolyEngine / PillEngine
Remote

MAINTAINER / ENGINE DEVELOPER — *Oct. 2017 – Present*

- Maintaining and evolving in-house and open-source engine codebases, including core runtime, architecture, and developer tooling.
- Designed and implemented engine subsystems and supporting data structures, with a strong focus on performance, iteration speed, and maintainability.
- Used these projects as a vehicle for exploring practical systems design, hot reloading, ECS-style architecture, and low-level engine programming.

KNTG Polygon
Warsaw, Poland

Vice President / Event Organiser — *Oct. 2017 – Jun. 2019*

- Co-led a student game development group, organised weekly meetings, and helped sustain a technical community around programming and game development.
- Delivered talks on modern C++ and practical language pitfalls, helping less experienced developers improve code quality and technical judgment.
- Planned and delivered community events, including Game Dev Fest and game jams, coordinating speakers, sponsorship, and logistics.

Selected Presentations

Everything Open
Canberra, ACT, Australia

Presenter – *Small Game Engines, Big Lessons: PolyEngine and Pill Engine* — *Jan. 2026*

Hackaday Supercon
Berlin, Germany

Presenter – *Hacking an FPGA Design Until It Breaks (and Fixing It Again)* — *Mar. 2025*

Embedded Open Source Summit
Seattle, WA, USA

Presenter – *ZLED Frame, or How to Illuminate Your Artwork* — *Apr. 2024*

FOSDEM
Brussels, Belgium

Presenter – *OpenRAN and Open Source: the Cool Kids of Telecom* — *Feb. 2024*

Hackaday Supercon; Embedded Open Source Summit
Pasadena, CA, USA; Prague, Czech Republic

Presenter – *Porting an AI-Powered Wearable Health Monitor to Zephyr on Open Hardware* — *Nov. 2023; Jun. 2023*

State of Open Con
London, United Kingdom

Presenter – *Hardware Acceleration Using FPGAs in Embedded Linux* — *Mar. 2023*

Honors & Awards

Dec. 2023	**Award in Data for City Hackathon 2023**, Urząd m.st. Warszawy	*Warsaw, Poland*
Nov. 2022	**Winner – Game Development Challenge**, HackYeah	*Kraków, Poland*
Aug. 2021	**Winner – Best Entrepreneurial Team**, EIT Digital Summer Health School	*Tallinn, Estonia*
Mar. 2018	**Honorable Mention**, Static Code Analysis Competition – Samsung Electronics	*Global*